

Algebra 1
Unit 9
Lesson 11 Practice

Name _____
Date _____
Period _____

A barber in Gainesville, Florida can track his expected revenue, $R(x)$, in dollars, each week based on what he charges for a basic haircut. The table shows the revenue model for the barber, based on x , the cost for a basic haircut.

Cost, x	2	4	6	8	10	12	14
Revenue, \$ $R(x)$	-100	2700	4700	5900	6300	5900	4700

1. Write a quadratic equation, in vertex form, that models the revenue, $R(x)$, in terms of the basic haircut's cost, x .
2. Write the quadratic equation in standard form.
3. Use the quadratic equation to determine how much the barber should charge for a basic haircut to estimate that the revenue coming that week is \$6075.

The table models the height, $h(x)$, in feet, after x seconds, of a toy projectile that is launched from 22 feet above the ground.

Seconds, x	1	2	3	4	5
Height, $h(x)$	85	130	157	166	157

4. Write a quadratic equation, in vertex form, that models the height of the toy in terms of seconds, x .

5. Rewrite the quadratic equation modeling the toy's height in standard form.
6. Use the quadratic equation to determine at what time the toy will land back on the ground.

The function, $M(x)$, models the number of miles, in millions, that U.S. passengers traveled on a train for the period from 1970 to 2000, where x is the number of years since 1970.

7. Write a quadratic equation, in vertex form, that models the miles traveled by U.S. train passengers, $M(x)$, in terms of years since 1970, x .

x	$M(x)$
14	4919
15	4904
16	4899
17	4904
18	4919
19	4944

8. Rewrite the quadratic equation in standard form.
9. Use the quadratic equation to determine when $y = 6179$ miles.
10. Interpret the solutions and determine if the solutions are viable in context.